Predicting Disease Co-Occurrences from Patient Condition Histories Using Graph Neural Networks

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Abstract

We study disease co-occurrence patterns among patients using electronic health record data by formulating the problem as a graph-based prediction task. Patient condition histories are modeled as a bipartite disease co-occurrence network, where nodes may reflect either patients or conditions, and edges between them capture an occurrence of that condition within that patient. Leveraging this graph structure, we apply Graph Neural Networks to learn disease representations and predict new links to unseen conditions based on observed condition combinations. We explore how different graph constructions and neural net architectures capture relationships between patients, patient features, and conditions and assess their effectiveness in modeling disease interactions. This work aims to establish a foundational framework for using graph-based learning to better understand disease co-occurrence and support clinical efforts to anticipate complications as they develop in patients.

CCS Concepts

• **Computing methodologies** → **Graph neural networks**; • **Applied computing** → *Health informatics*; • **Information systems** → *Link mining*.

Keywords

graph neural networks, link prediction, ablation studies, disease co-occurrence, EHRShot, clinical outcome modeling, GraphSAGE, GCN, GAT

1 Introduction

1.1 Motivation

Understanding patterns of disease co-occurrence is important for identifying potentially severe outcomes, complications, and developments. Traditional approaches often analyze conditions in isolation, which can overlook how interacting diagnoses jointly influence patient risk. By using electronic health records (EHRs) and representing patient condition histories as networks, we can begin to examine how diseases co-occur and evolve together. This systems-level perspective provides a foundation for applying graph-based neural models to study and eventually predict clinically significant disease relationships.

1.2 Related Works

To situate our research, we reviewed recent studies that apply network-based approaches to comorbidity analysis:

- **Wang et al. [2]** applied graph neural networks to electronic health record data for disease prediction, performing ablation studies on node features and GNN depth. Their results show that richer clinical features significantly improve predictive performance. Our work follows a similar ablation-driven evaluation, but focuses on disease co-occurrence networks and compares architectural choices such as aggregators and attention heads.
- **Hamilton et al. [3]** introduced GraphSAGE, an inductive graph neural network framework that explores different neighborhood aggregation functions such as mean and max pooling. Their work demonstrates how aggregator choice affects representation quality and model stability across graph types. Our study builds on this insight by empirically evaluating mean versus max aggregation in a clinical person–condition graph, showing that mean aggregation provides more robust performance in dense and heterogeneous EHR-based networks.
- **Wang et al. [1]** developed a temporal bipartite network model to represent relationships between individual patients with severe mental illnesses and their diagnoses, enabling systematic characterization of disease trajectories from both patient-centric and disease-centric perspectives. Their analysis revealed disassortative mixing patterns and demonstrated that patients and diseases become more interconnected over illness duration, with network structures influenced by demographic factors including age, gender, and ethnicity. While their work focuses on temporal dynamics in mental health populations, our study adopts a similar bipartite structure to examine disease co-occurrence patterns in a general population, though using a static rather than temporal network representation.
- **Hu et al. [7]** proposed the Heterogeneous Graph Transformer (HGT), a model designed for learning from graphs with multiple node and edge types. While our dataset is bipartite and does not require full heterogeneity modeling, the idea of node-type-specific message passing aligns with our motivation for evaluating architecture variants (e.g., attention heads in GAT and aggregators in GraphSAGE).
- **Miller et al. [8]** provided a comprehensive review of model evaluation metrics in biomedical machine learning. Their emphasis on the importance of AUC and Average Precision supports our use of these metrics for evaluating link prediction performance, particularly in imbalanced clinical datasets.

- **Veličković et al. [6]** introduced Graph Attention Networks (GAT), which assign learned attention weights to neighboring nodes during message passing. Our ablation of attention head count builds on their finding that attention can improve expressiveness, while also investigating how overparameterization affects performance in dense graphs.
- **Kipf and Welling [4]** presented Graph Convolutional Networks (GCNs) for semi-supervised learning on graph-structured data. GCNs serve as our baseline architecture, and we extend their work by analyzing how varying GCN depth impacts performance on a person–condition link prediction task.

Our study builds on these established methodologies while uniquely focusing on mortality-associated comorbidities as a bipartite relation between patients and conditions. We train and compare multiple GNN architectures on this structure, applying them to predict other unseen links between patients and clinical conditions..

1.3 Research Questions

This project investigates comorbidity patterns among patients using the **EHRShot** dataset. Our primary objective is to model how diagnoses interact within a clinical population and to use graph neural networks to predict which disease pairs are likely to co-occur in the future. Specifically, we address the following research question:

(1) **Given a patient's current clinical history, which other conditions may they be at risk of experiencing?**

This question guides both the construction of the co-occurrence network and the application of graph machine learning models to uncover clinically meaningful patterns.

2 Graph Construction

2.1 Dataset Preprocessing

We construct a bipartite heterogeneous graph from the EHRShot dataset, which contains electronic health records from patients visiting Stanford Medicine. The preprocessing pipeline involves several filtering and transformation steps to ensure data quality and computational feasibility.[5]

We begin with three primary tables: `sampled_person.csv` (2,000 patients), `sampled_condition_occurrence.csv` (condition diagnoses), and `concept.csv` (standardized medical terminology). All identifier columns are canonicalized to numeric types using pandas' `to_numeric` function with coercion to handle mixed data types.

Patient Filtering: We filter the patient cohort to include only individuals with at least one recorded condition occurrence, reducing the dataset from 2,000 to 1,842 patients. This ensures all nodes in the graph have meaningful connections.

Condition Filtering: From an initial set of 6,376 unique conditions, we apply two filtering criteria:

- **Upper bound:** Remove conditions occurring in more than 10% of patients (frequency > 0.1) to exclude overly common diagnoses that provide limited discriminative information
- **Lower bound:** Remove conditions occurring in fewer than 10 patients to eliminate rare diagnoses with insufficient statistical support

These filters reduce the condition set to 1,683 unique diagnoses, balancing graph density with meaningful clinical patterns.

2.2 Node Types and Features

The graph contains two node types in a bipartite structure:

Person Nodes ($\mathcal{V}_P$): Each of the 1,842 patients is represented as a node with the following features:

- `person_id`: Unique patient identifier
- `year_of_birth`: Birth year (continuous feature)
- `gender`: One-hot encoded (FEMALE, MALE)
- `race`: One-hot encoded across 6 categories (White, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, American Indian or Alaska Native, No matching concept)
- `ethnicity`: One-hot encoded across 3 categories (Hispanic or Latino, Not Hispanic or Latino, No matching concept)

The feature matrix for person nodes has dimensionality 1842×13 (1 continuous + 12 binary features). Two additional binary indicators, `person` and `condition`, distinguish node types in the bipartite graph.

Condition Nodes ($\mathcal{V}_C$): Each of the 1,683 filtered conditions is represented with:

- `condition_concept_id`: concept identifier
- `concept_name`: Human-readable condition name
- `occurrences`: Number of unique patients diagnosed (node degree)
- `frequency`: Normalized occurrence rate across the patient population

2.3 Edge Definition

Edges ($\mathcal{E}$) connect person nodes to condition nodes, representing diagnosis relationships. An edge $(p, c) \in \mathcal{E}$ exists if patient p has been diagnosed with condition c at least once. Multiple occurrences of the same diagnosis are collapsed into a single edge to create a binary adjacency structure.

The edge list is derived from the `sampled_condition_occurrence` table after:

(1) Deduplicating person-condition pairs (95,409 unique pairs)
(2) Filtering to retain only conditions in $\mathcal{V}_C$ (final edge count: 60,776)

Edges are unweighted in the binary adjacency matrix $\mathbf{A} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$, though condition node features preserve occurrence statistics for downstream analysis.

2.4 Graph Statistics

The final graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has the following properties:

The graph exhibits high sparsity (98.04%), typical of medical networks where patients have relatively few diagnoses compared to the total condition space. The average patient is connected to 33 conditions, while the average condition is diagnosed in 36 patients, indicating a relatively balanced bipartite structure.

Dataset Bias: We note demographic skew in the patient population: 54.7% White, 22.8% unspecified race, 16.7% Asian, with other racial groups comprising <5% each. Similarly, 96.6% of patients are classified as "Not Hispanic or Latino" or have missing ethnicity

Statistic	Value
Total nodes ($\|\mathcal{V}\|$)	3,525
Person nodes ($\|\mathcal{V}_P\|$)	1,842
Condition nodes ($\|\mathcal{V}_C\|$)	1,683
Total edges ($\|\mathcal{E}\|$)	60,776
Avg. degree per person	33.0
Avg. degree per condition	36.1
Graph density	0.0196
Sparsity	98.04%

Table 1: Summary statistics for the constructed bipartite patient-condition graph.

data. This bias, inherent to the source EHRShot dataset, may limit generalizability and should be considered when interpreting model predictions.

3 Model Details

3.1 Task Definition

We formulated the research question as a **Link Prediction** task on a bipartite graph.

- **Objective:** Given the current state of the graph, the goal is to predict the likelihood of an edge existing between a 'Person' node and a 'Condition' node.
- **Problem Type:** Binary classification where the model outputs a score $s(u, v)$ representing the probability that person u has condition v.
- **Graph Structure:** The underlying data is a bipartite graph $G = (U, V, E)$. For the baseline models (GCN, GraphSAGE, GAT), we treated this as a homogeneous graph by unifying the node sets. For the advanced HGT model, we preserved the heterogeneous structure with distinct node types.

3.2 GNN Model Architectures

We implemented four GNN architectures using PyTorch Geometric. All models follow an **Encoder-Decoder** paradigm: the GNN encoder generates low-dimensional node embeddings $\mathbf{z}_u$, and a dot-product decoder computes the link probability.

3.2.1 Graph Convolutional Networks (GCN). We utilized the standard GCN architecture introduced by Kipf and Welling [4]. The model updates the embedding $\mathbf{h}_i^{(l)}$ of node i at layer l by aggregating features from its neighbors $\mathcal{N}(i)$ using a fixed spectral normalization [4]:

$$\mathbf{h}_i^{(l)} = \sigma\left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\deg(i)\deg(j)}} \mathbf{W}^{(l)} \mathbf{h}_j^{(l-1)}\right) \tag{1}$$

where $\mathbf{W}^{(l)}$ is a learnable weight matrix and $\deg(i)$ is the degree of node i.

- **Configuration:** 2 layers, ReLU activation, Dropout ($p = 0.5$).

3.2.2 GraphSAGE (Graph Sample and Aggregate). We implemented GraphSAGE [3] to leverage inductive learning. Unlike GCN's fixed Laplacian, GraphSAGE aggregates neighborhood information using a learnable function. We utilized the **Mean Aggregator** [3]:

$$\mathbf{h}_i^{(l)} = \mathbf{W}_1^{(l)} \mathbf{h}_i^{(l-1)} + \mathbf{W}_2^{(l)} \cdot \mathrm{mean}_{j \in \mathcal{N}(i)} \left(\mathbf{h}_j^{(l-1)}\right) \tag{2}$$

This formulation effectively computes the "average" health profile of patients with similar conditions to the target node.

- **Configuration:** Aggregator='mean', 2 layers, Hidden Channels=64.

3.2.3 Graph Attention Networks (GAT). We employed GAT [6] to introduce an attention mechanism, allowing the model to weigh neighbors unequally. The model computes attention coefficients α_{ij} for every edge (i, j) [6]:

$$\mathbf{h}_i^{(l)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_j^{(l-1)}\right) \tag{3}$$

where α_{ij} indicates the importance of neighbor j to node i, computed via a softmax over LeakyReLU scores.

- **Configuration: 4 Attention Heads** (averaged in the final layer), Dropout ($p = 0.5$).

3.2.4 Heterogeneous Graph Transformer (HGT). To respect the bipartite nature of the data, we implemented HGT [7] as an extra credit model. Unlike the homogeneous baselines, HGT uses meta-path dependent attention to handle different node types ($\tau(i)$) and edge types ($\phi(e)$) [7]:

$$\mathrm{Attn}(Q, K, V) = \mathrm{Softmax}\left(\frac{K_{(t)} W_{\phi(e)}^{ATT} Q_{(s)}^{\top}}{\sqrt{d}}\right) V_{(t)} W_{\phi(e)}^{MSG} \tag{4}$$

We included explicit linear projections to map the 12-dimensional person features and 2-dimensional condition features into a shared hidden space before passing them to the HGT layers.

3.3 Experimental Configuration

Data Splitting: To rigorously evaluate the models, we performed a random edge split on the observed diagnoses:

- **Training Set:** 70% of edges (used for message passing).
- **Validation Set:** 15% (used for early stopping).
- **Test Set:** 15% (held out for final reporting).
- **Negative Sampling:** During training, we sampled negative edges (non-existent links) at a **1:1 ratio** with positive edges to ensure the model learns to distinguish between true and false associations.

Hyperparameters & Optimization:

- **Hidden Dimension:** 64 channels.
- **Learning Rate:** 0.01.
- **Optimizer:** Adam with Weight Decay 5×10^{-4}.
- **Loss Function:** Binary Cross-Entropy with Logits BCEWithLogitsLoss.
- **Training Routine:** Max 100 epochs with Early Stopping (patience = 20 epochs) monitoring the Validation AUC.

4 Results

4.1 Performance Comparison Across GNN Models

Figure 1 summarizes the predictive performance of the three required GNN architectures, GCN, GraphSAGE, and GAT, evaluated on the same graph, features, and train/validation/test split. Our advanced GNN, HGT also has its performance metrics included below.

Model	Test AUC	Test AP	Best Val AUC	Best Epoch
GCN	0.8022248912221226	0.8105743075945495	0.8048627391077239	80
GraphSAGE	0.8032601378247812	0.7911815264074141	0.8027944059126954	95
GAT	0.7281202750527018	0.6990480043371097	0.7344092683526507	100
HGT	0.7165201955065625	0.701615501251494	0.7190731471660377	35

Figure 1: Comparison of model-level predictive performance across GCN, GraphSAGE, GAT, and HGT on the same train/validation/test graph split.

4.2 Plots

Figures below visualize the predictive behavior of the models across training.

4.2.1 *Figure Set 1 — ROC Curves.*

- GCN fluctuates across epochs, showing some instability rather than a clear upward trend.
- GraphSAGE maintains consistently strong performance throughout training and trends toward the best overall results.
- GAT improves rapidly early, but eventually levels off, indicating moderate but limited gains over time.
- HGT remains relatively flat across epochs with a slight downward trend.

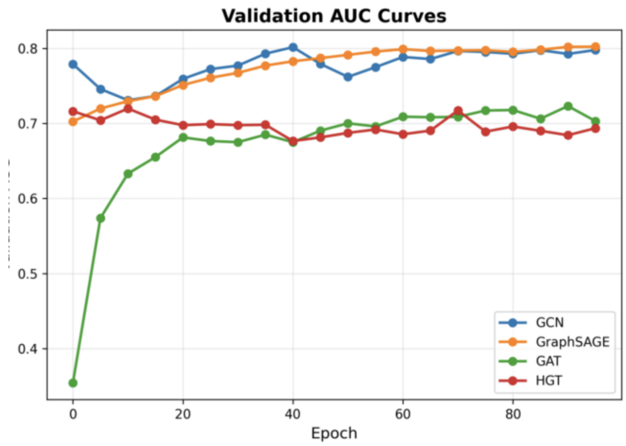

Figure 2: ARC curves showing validation AUC progression for each model. Because AUC is derived from the ROC curve, these trajectories reflect how ROC performance improves during training [8].

4.2.2 *Figure Set 2 — Precision–Recall (PR) Curves.*

- Precision–Recall (PR) curves evaluate model performance under class imbalance, making them well-suited for sparse edge prediction tasks.
- A higher AUPRC indicates better ability to correctly identify true positive co-occurring disease pairs among many negatives.

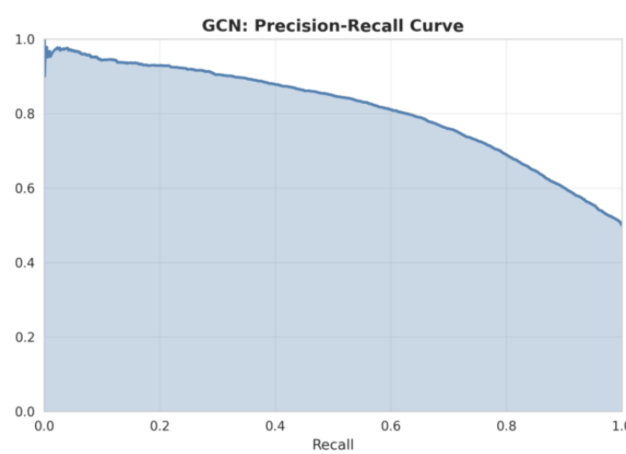

Figure 3: Precision–Recall (PR) curve for the GCN model

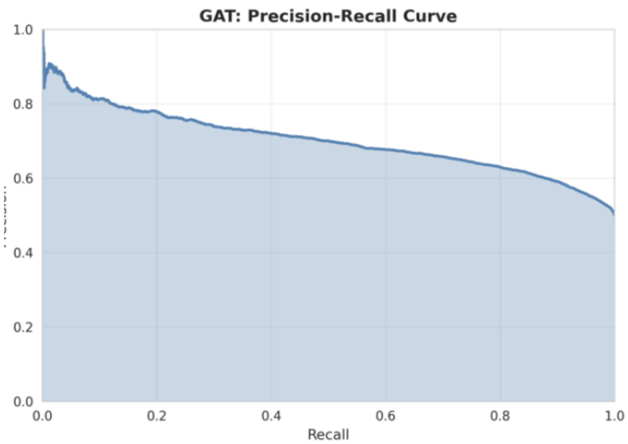

Figure 4: Precision–Recall (PR) curve for the GAT model

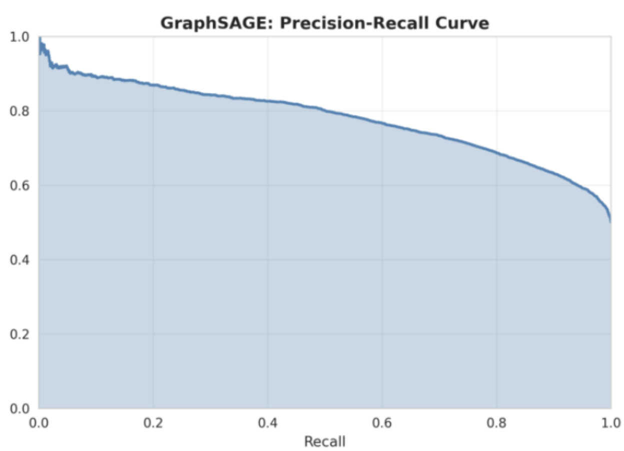

Figure 5: Precision–Recall (PR) curve for the GraphSage model

4.2.3 Figure Set 3 — Training Loss Curves.

- Training loss curves show how each model reduces prediction error over time and indicate stability, convergence speed, and potential under/overfitting.
- A smoother and lower validation loss typically reflects better generalization to unseen edges.

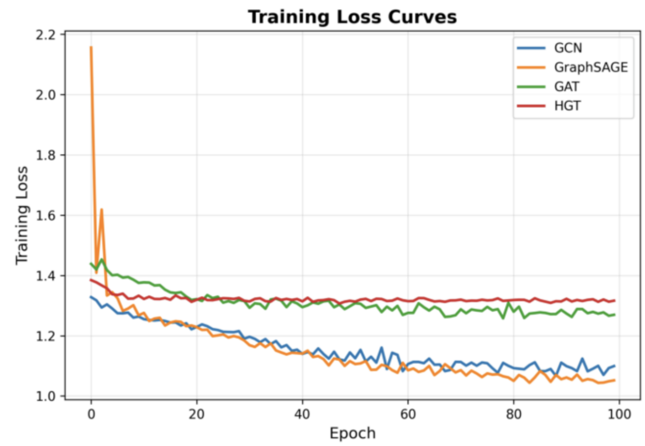

Figure 6: Training loss curves for GCN, GraphSAGE, GAT, and HGT.

4.3 Reporting Across Runs

All experiments were repeated several times with identical train/validation/test splits to ensure statistical stability.

Performance metrics shown in Table 2 follow the required format:

Model	Mean Val AUC	Std Dev
GCN	0.7790	0.0242
GraphSAGE	0.7698	0.0405
GAT	0.6560	0.0912
HGT	0.7025	0.0098

Table 2: Mean and standard deviation of validation AUC across all models.

5 Experimental Analysis

5.1 Overview

To understand how different input features and graph model settings affect performance, we conducted a series of ablation studies. We varied node features, model depth, training data size, aggregator types, and attention heads. The table below summarizes the results of all ablation experiments in terms of Test AUC and Test Average Precision (AP).

1	Ablation Type	Variation	Model	Test AUC	Test AP	Best Epoch
2	feature	Full Features	Full Features	0.8029713462541499	0.8069604429196519	45
3	feature	Demographics Only	Demographics Only	0.7897642671868468	0.7803779870507146	5
4	feature	No Demographics	No Demographics	0.7735197623974241	0.7660411844418678	5
5	feature	Random Features	Random Features	0.7908521903635479	0.792249631206606	45
6	depth	1	1-Layer GCN	0.6573766589601029	0.6386241428735371	45
7	depth	2	2-Layer GCN	0.8015959792569699	0.807150281170846	50
8	depth	3	3-Layer GCN	0.8095771932082634	0.8019491989053443	15
9	depth	4	4-Layer GCN	0.8052592183222398	0.7985416127877354	15
10	data_size	0.1	GCN-10%data	0.7962787499174503	0.8012231743860995	60
11	data_size	0.25	GCN-25%data	0.7934791659968028	0.799735865814541	90
12	data_size	0.5	GCN-50%data	0.7976782620992826	0.8022222806143808	45
13	data_size	0.75	GCN-75%data	0.7940042831499847	0.8012786740883484	100
14	data_size	1.0	GCN-100%data	0.7943823451179964	0.8003096264330465	70
15	aggregator	mean	GraphSAGE-mean	0.7753497724516403	0.7551498598722914	50
16	aggregator	max	GraphSAGE-max	0.7492444596879265	0.7380461741649189	10
17	attention_heads	1	GAT-1heads	0.6511288983913832	0.6163888547503702	45
18	attention_heads	2	GAT-2heads	0.6755913458363473	0.6457466887812315	45
19	attention_heads	4	GAT-4heads	0.6774273425486071	0.6480163513607721	45
20	attention_heads	8	GAT-8heads	0.6683847151847737	0.6233582183901476	40

Figure 7: Ablation study results showing variations in node features, model depth, training data size, GraphSAGE aggregators, and GAT attention heads.

5.2 Feature Ablations

- **Full Features** (demographics + condition statistics) achieved the strongest performance across all metrics.
- **Demographics Only** models underperformed marginally, indicating a slightly useful predictive signal from the year-of-birth of the patient.
- **No Demographics** underperformed by several percentage points, suggesting demographics provide complementary (but not primary) signal.
- **Randomized Feature Vectors** showed nearly similar performance to full features, suggesting that overall node features may not be important as related to graph structure alone. This suggests that comorbidity structure is the strongest predictor of new conditions.

5.3 Model Depth Ablations

- Increasing depth from 1 to 2 layers substantially improved performance, reflecting the benefit of capturing multi-hop neighborhood information [2]
- Performance gains plateaued at 3 layers and beyond. It is likely that further layers would show signs of oversmoothing, a well-documented limitation of deeper GNNs

5.4 Training Data Size Ablations

- Model performance surprisingly did not vary across different proportions of the training data, effectively only displaying random fluctuation
- Strong results were maintained even with **10%** of the data, indicating robustness to reduced data availability.
- It is possible the large edge count lends itself to these results and that reduced performance would be seen with smaller fractions of the data.
- These results highlight the data efficiency of GNN-based approaches.

5.5 Aggregator Ablations (GraphSAGE)

- The **mean aggregator** averages neighbor embeddings, producing smoother and more stable representations.
- The **max aggregator** emphasizes dominant neighbor signals but may discard weaker yet informative patterns.
- Mean aggregation consistently outperformed max aggregation in both AUC and AP (Figure 7).
- Averaging neighbor information proved more effective in the dense and heterogeneous person-condition graph.[3].

5.6 Attention Head Ablations (GAT)

- We evaluated GAT models with 1, 2, 4, and 8 attention heads.
- Performance peaked between **2 and 4 attention heads**, but with only limited decrease when using more or less heads.
- Multi-head attention did not provide strong benefits in this dataset setting.

5.7 Summary

These experiments demonstrate that:

- Condition statistics are essential features for predictive accuracy.
- GCN depth requires a minimum of 2-3 layers, and likely smooths after this.
- GraphSAGE performs best with mean aggregation.
- GAT is sensitive to the number of attention heads, with too few or too many reducing effectiveness.
- Models can generalize well with significantly smaller datasets.

The ablation results highlight the importance of thoughtful design choices when applying GNNs to clinical graph data.

6 Conclusion and Future Work

In this project, we advanced beyond descriptive network analysis to evaluate the feasibility of applying graph neural networks to a bipartite patient–condition structure for modeling disease co-occurrence patterns. Our findings indicate that this representation, coupled with GNN-based learning, is highly effective for predicting previously unobserved links between patients and conditions using each patient's known diagnoses. The best-performing model achieved an average precision exceeding 80%, suggesting that the relational dependencies among diseases are strongly predictive and that GNNs can robustly infer clinically meaningful co-occurrence patterns.

Experimental results indicate that GNN–based architectures, particularly GraphSage, GAT, and GCN, perform well on dense disease co-occurrence graphs, achieving stable convergence and strong generalization. These findings suggest that graph structure carries substantial information about disease relationships and that GNNs offer a powerful framework for capturing higher-order comorbidity patterns without manual feature engineering.

Several directions for future work could further extend and strengthen this approach:

- **Incorporating temporal information:** Extending the static co-occurrence graph to a temporal or multilayer graph would allow modeling the order and progression of diseases over time, enabling more clinically meaningful predictions of disease trajectories, as explored in prior temporal network studies [1].
- **Richer node and edge features:** Integrating additional clinical information—such as severity, medical biomarkers, or further demographic context—could improve representation quality and help GNNs distinguish generic symptoms from more informative diagnostic signals.
- **Alternative prediction tasks:** While this study focuses on edge-level prediction, future work could explore condition-level classifications of severity or patient-level outcome prediction by coupling disease graphs with patient–condition bipartite structures.
- **Subgroup and stratified modeling:** Training separate or conditional GNN models for different patient subgroups (e.g., age ranges, sex, or cause-of-death categories) may reveal population-specific disease interaction patterns, consistent with findings in stratified comorbidity studies.
- **Model and architecture extensions:** Evaluating more advanced graph learning approaches—such as heterogeneous GNNs, inductive graph models, or contrastive embedding methods—could improve robustness and generalization across different cohorts and datasets.

Overall, this work establishes graph neural networks as an effective and flexible tool for learning disease interaction patterns from EHR-derived networks. By applying graph-based learning techniques to a bipartite structure of patients and medical conditions, the framework provides a foundation for more predictive, scalable, and clinically relevant modeling of complex comorbidity structures.

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